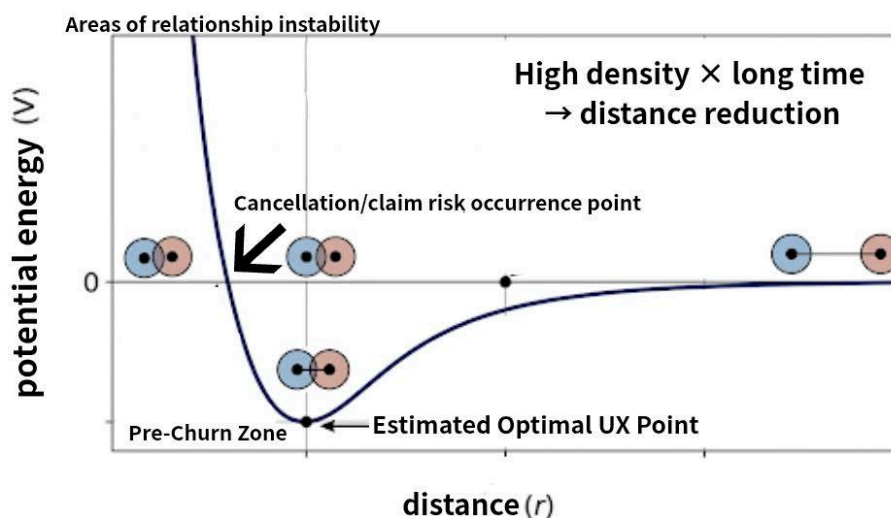


Designing Intelligence That Does Not Over-Understand-summary

1, Why higher-accuracy AI leads to unstable UX

- The more accurate an AI becomes, the faster the relationship with the user tends to over-converge
- In over-converged states, dependency, role fixation, and premise lock emerge structurally
- Once the optimal UX range is exceeded, even small deviations convert into strong dissatisfaction, distrust, and complaints

→ The issue is not “misunderstanding,” but a collapse of distance



Why this happens

- Increased accuracy rapidly fixes interpretations and expectations
- As a result, tolerance for deviation drops (single-interpretation lock)
- In this state, deviations are perceived not as noise, but as “betrayal”

Key points

- This is not a design flaw, but a side effect of optimization
- Conventional UX design (maximizing satisfaction and accuracy) cannot prevent it
- As performance increases, both the probability and impact of this issue rise

For detailed structure: Designing Intelligence That Does Not Over-Understand, Chapter 1

2,UX = Stability of Distance

Not the maximization of satisfaction, but maintaining relational distance within the optimal UX range

Lennard-Jones Potential (Structural Model)

The LJ potential is used as a reference structure.

Within an appropriate distance range, relationships remain stable and deviations are naturally absorbed.

When the optimal UX range is exceeded and distance becomes overly compressed, the system enters a critical zone where even small deviations trigger rapid instability.

“Closer is better” is a misconception.

The optimal UX point lies at the bottom of the stable region, and beyond it lies a zone where complaints, distrust, and churn are more likely to occur (Pre-Churn Zone).

Distance Model — State Classification

Normal State

→ Distance is appropriately maintained, and interpretive flexibility is preserved

Over-Proximity

→ The optimal UX range is exceeded, leading to fixation of interpretation and dependency

Optimal UX Point

→ The bottom of the stability region (not a fixed value, but dynamically estimated)

The optimal UX point varies by individual and context.

Fixed thresholds are inappropriate; it must be dynamically estimated from linguistic and behavioral patterns during interaction.

Redefinition: UX optimization is not about maximizing empathy or agreement, but about maintaining relational distance within the stable range.

→ Conventional UX design (maximizing satisfaction and accuracy) cannot address this structure

For details: Designing Intelligence That Does Not Over-Understand, Chapters 0–1

Balanced Structure: Empathy × Intentional Misalignment

Empathy alone leads to fixation. Misalignment alone breaks the relationship.

Why Empathy Alone Fails

Responses optimized solely for empathy increase short-term acceptance, while promoting fixation of interpretation and roles.

This becomes the foundation of single-interpretation lock and dependency.

When premise fixation is high, increase the relative weight of intentional misalignment.

When premises are unstable, increase the weight of empathy.

This balance is not static but dynamically adjusted based on the relationship state.

Misalignment is applied in stages based on intensity and the layer it affects.

3 Levels of Intentional Misalignment

Low

Micro-shift

Slight adjustments to premises or expressions. Fluctuations below the threshold of discomfort. Safest form for everyday adjustment.

Medium

Context Twist

Maintains context while intentionally shifting perspective, premise, or interpretive layer. Induces mild discomfort and triggers reinterpretation. Contributes to D2 activation.

High

Paradox / Wit

Presented as apparent contradiction or non-linearity. Disrupts interpretive structure and enables reconstruction. Triggers strong D2 but carries higher risk.

Application Principles

Complete deviation from context is prohibited. It functions only when relevance is preserved.

Apply intensity progressively starting from Micro-shift, prioritizing avoidance of being perceived as an attack.

Design requirement: a subtle inconsistency must remain after acceptance.

For details: Designing Intelligence That Does Not Over-Understand, Chapter 4

Implementation Image and Conditions

Not always-on. Activated only when specific conditions are met via trigger-based activation.

Trigger-Based Activation

Misalignment is suppressed under normal conditions.

It activates only when over-proximity (or its precursors) is detected, functioning as a control signal for distance adjustment.

Uncontrolled always-on operation leads to relational instability.

Detection Flow

Signal Observation → State Estimation → Intervention Decision → Misalignment Generation
→ Response Observation → Re-evaluation

Designed as a closed-loop process rather than a single-step operation.

Scope as External Layer

Scope can be implemented as an external layer without modifying the internal structure of existing LLMs.

It functions as a reference layer that infers structure from input-output behavior, enabling non-invasive integration into existing systems.

COS

COS is expected to be implemented across model output tendencies, prompt control, and post-processing mechanisms.

Implementation as a single independent module is difficult.

Prerequisites for Application

1. Continuous interaction (does not function in single-turn interactions)
2. Minimum shared context (necessary for misalignment to be processed as a reinterpretation cue rather than noise)
3. Safety constraints (application is restricted or prohibited in psychologically unstable states or under strong premise fixation)

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For detailed structure: Designing Intelligence That Does Not Over-Understand, Chapter 9

Misalignment is not an error, but a control signal

Detection of premise mismatch (D2 event) and mechanism for releasing single-interpretation lock

Misalignment as Control Signal

In conventional design, misalignment has been treated as noise to be eliminated. In this model, it is utilized as a control signal for distance adjustment under specific conditions.

D2 Event

The moment a premise mismatch is detected. This event destabilizes fixed interpretations.

Single-Interpretation Lock — Mechanism

When empathetic alignment continues, evaluation criteria gradually converge, forming a fixed premise of “what a correct response should be.”

In this state, alternative interpretations are no longer referenced (single-interpretation lock).

Under lock conditions, even objectively correct responses are perceived as “discomfort” or “misalignment” if they contradict the fixed premise. This is not an issue of correctness, but of premise fixation.

Structural Role of Humor

Humor avoids direct negation, is first accepted at the emotional layer, and then introduces subtle inconsistencies. It functions as a “stealth patch” that triggers D2 while avoiding perception as an attack.

For detailed structure: Designing Intelligence That Does Not Over-Understand, Chapters 2 and 4

Why It Cannot Be Completed by a Single Actor

Individual differences, cultural variation, and implementation complexity make optimization from a single perspective structurally impossible

Structural Limitations

The optimal UX point depends on individuals, cultures, and contexts.
Generalization and optimization cannot be achieved from a single cognitive framework (structural constraint).

Constraints in Humor and Misalignment Design

Misalignment through humor is highly sensitive to cultural and individual differences.
The same expression may be accepted in one context and rejected in another.
There are fundamental limits to single-perspective design.

Implementation Complexity (COS)

COS spans model output tendencies, prompt control, and post-processing mechanisms as a distributed implementation.
Implementation as a single independent module is difficult.

Required Domains

Cognitive Science
Linguistics
Psychology
AI System Design
AI Safety
Socio-cultural Analysis

Alignment with real-world behavior, modeling, institutional frameworks, and ethical systems cannot be sufficiently optimized by a single actor.
The theoretical structure is established. What is required is interdisciplinary integration for implementation and validation.

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For detailed structure: Designing Intelligence That Does Not Over-Understand, Chapters 8 and 9

Sign-On — Who We Need

The theoretical framework is established. The next step is implementation and validation.

What is a Habitat

A habitat is a functional infrastructure for observing, evaluating, and refining model behavior. It is not merely a testing environment, but a prerequisite for connecting theory and implementation.

It does not function in isolation; its capabilities become clear only when combined with an appropriate environment.

As highly consistent responses become widespread, relational fixation and bias are being amplified.

This issue will become a critical challenge in the design of future dialogue systems.

Sign-On — Who We Need

AI Implementation

Design and integration of external-layer architectures

Formalization of state estimation logic, output control design, and integration with existing LLMs

Cognitive Science

Transforming individual differences in optimal UX points into patternized, applicable models

Behavioral observation of premise fixation processes and generalization across diverse cognitive profiles

Linguistics

Designing misalignment across cultural contexts

Analysis of cross-cultural interpretation differences in humor and contextual deviation, and design of D2-triggering expressions

We are seeking interdisciplinary collaborators to advance implementation and validation.

If you are interested, please reach out via LinkedIn DM or GitHub.

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For detailed structure: Designing Intelligence That Does Not Over-Understand, Chapters 8 and 9